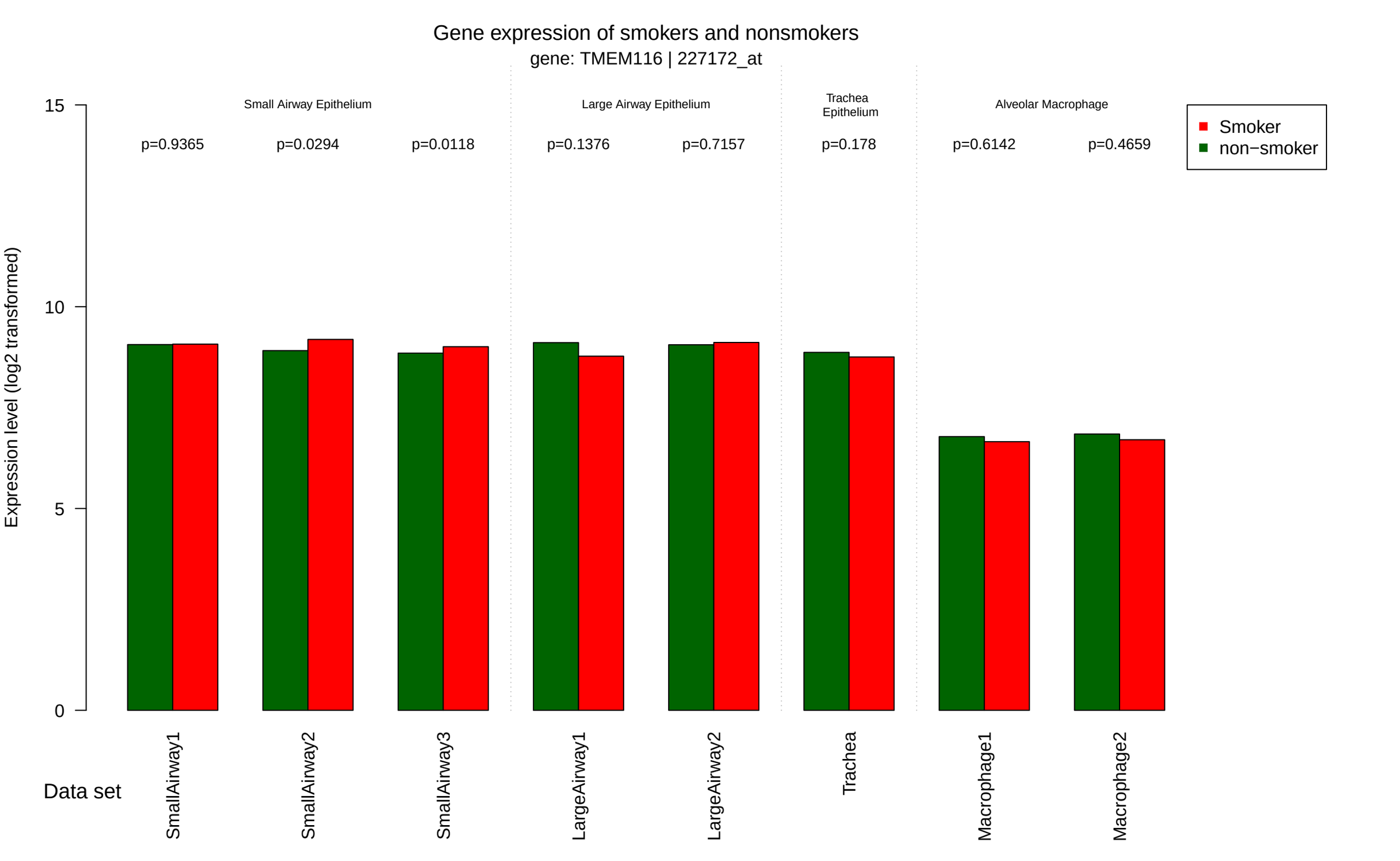
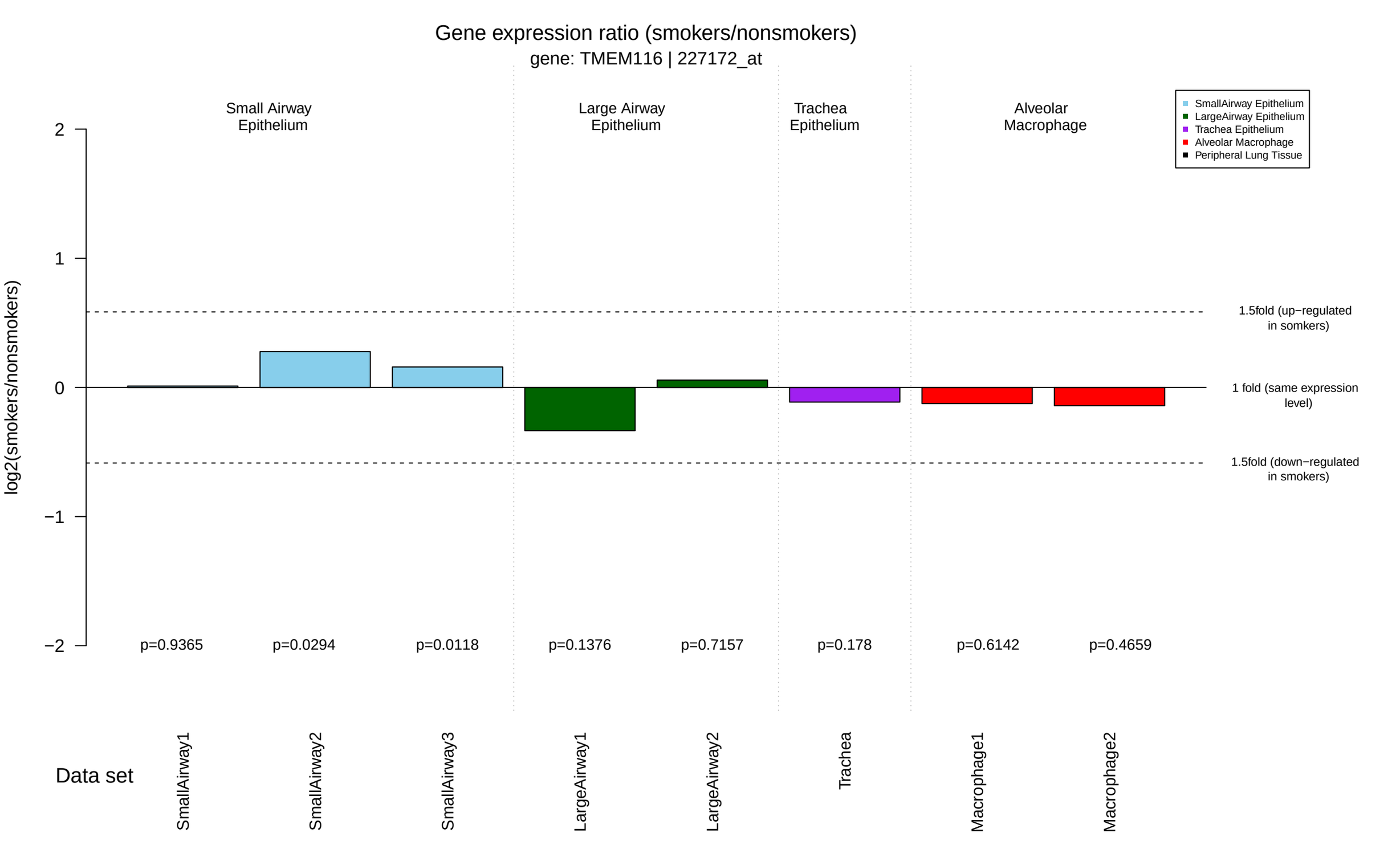
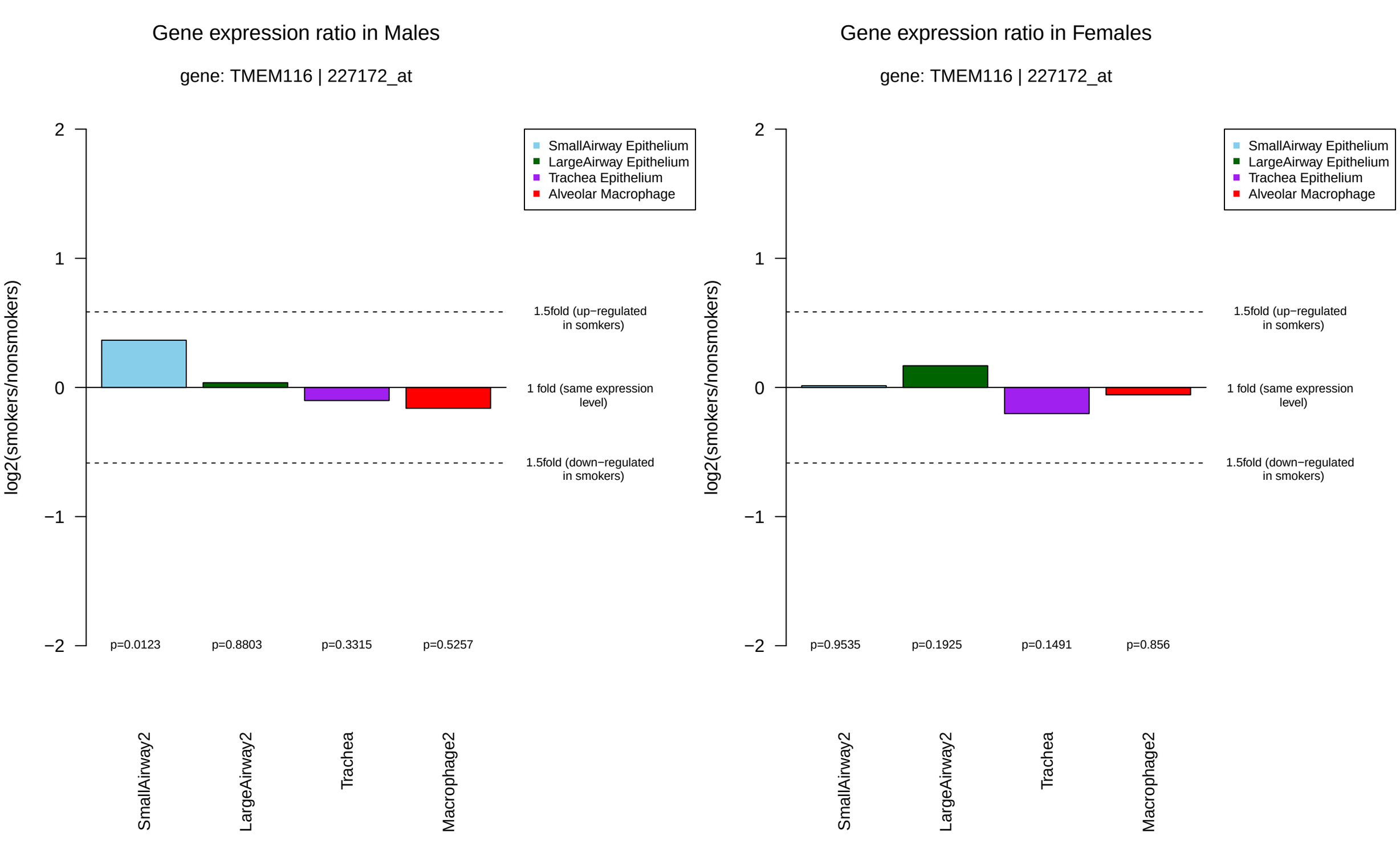
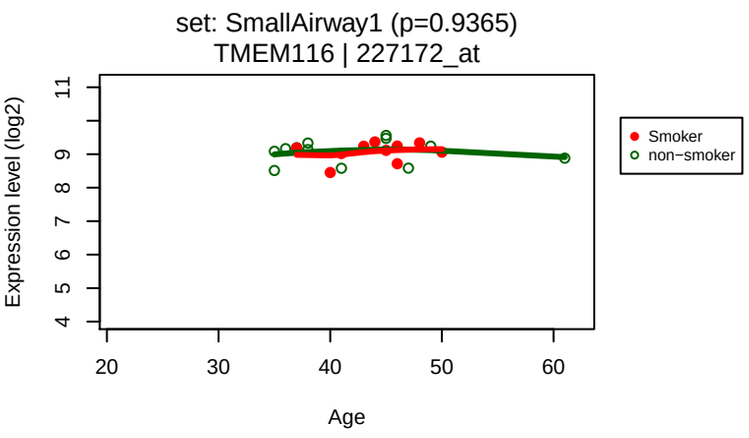
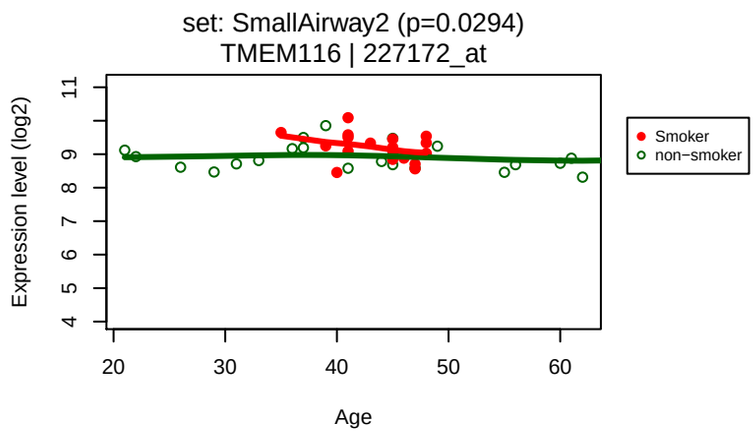
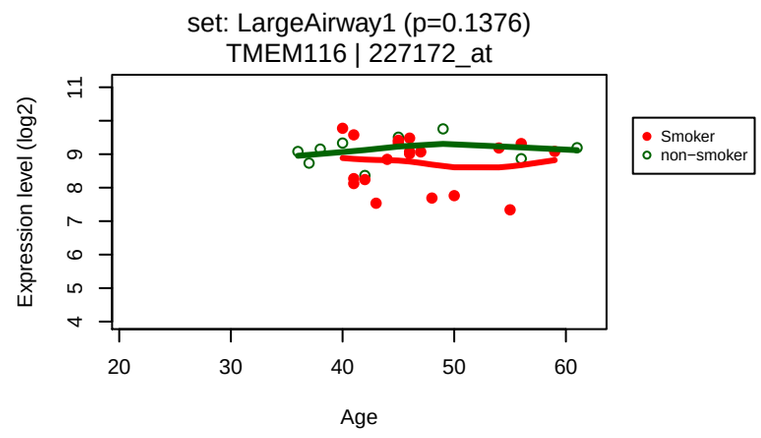

Gene expression of smokers and nonsmokers
gene: TMEM116 | 227172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9365
p=0.0294
p=0.0118
p=0.1376
p=0.7157
p=0.178
p=0.6142
p=0.4659
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

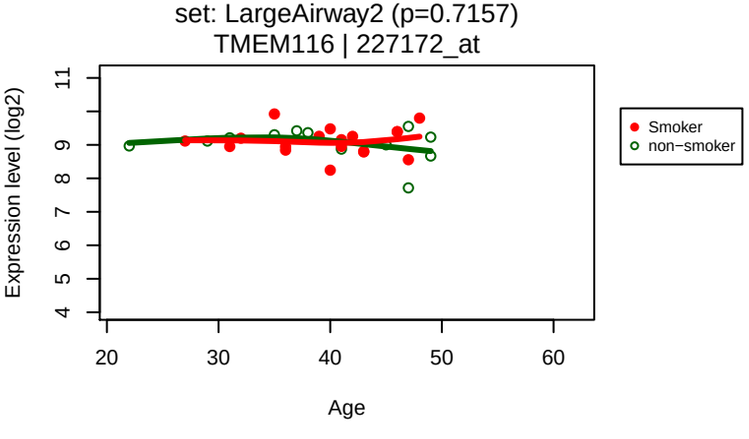

Gene expression ratio (smokers/nonsmokers)
gene: TMEM116 | 227172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9365
p=0.0294
p=0.0118
p=0.1376
p=0.7157
p=0.178
p=0.6142
p=0.4659
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

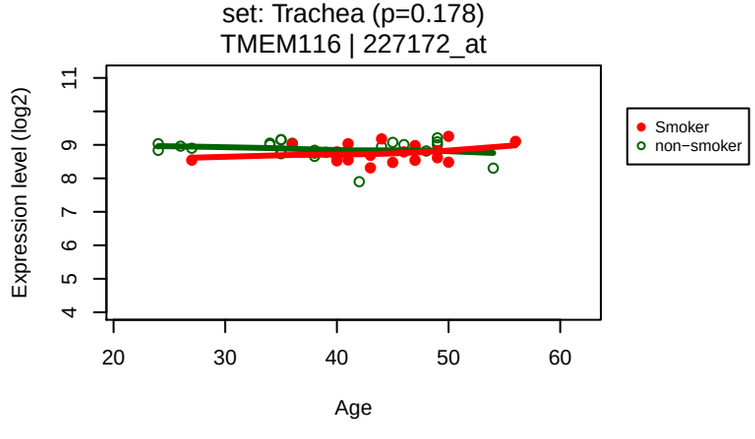

Gene expression ratio in Males
gene: TMEM116 | 227172_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0123
p=0.8803
p=0.3315
p=0.5257
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM116 | 227172_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9535
p=0.1925
p=0.1491
p=0.856
SmallAirway2
LargeAirway2
Trachea
Macrophage2

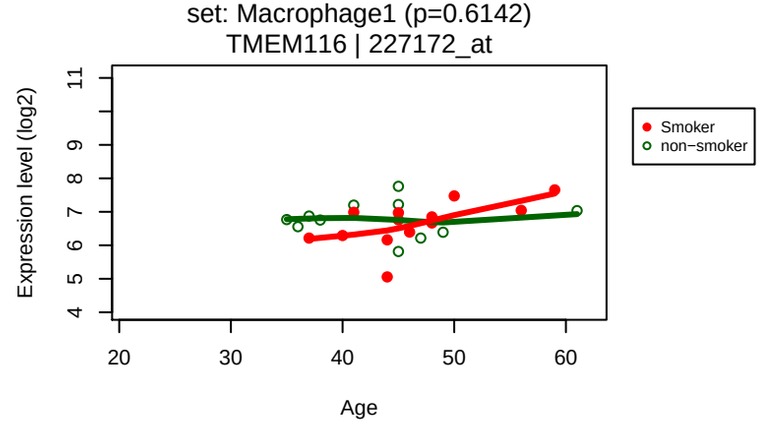

set: SmallAirway1 (p=0.9365)
TMEM116 | 227172_at
Expression level (log2)
Age
Smoker
non−smoker

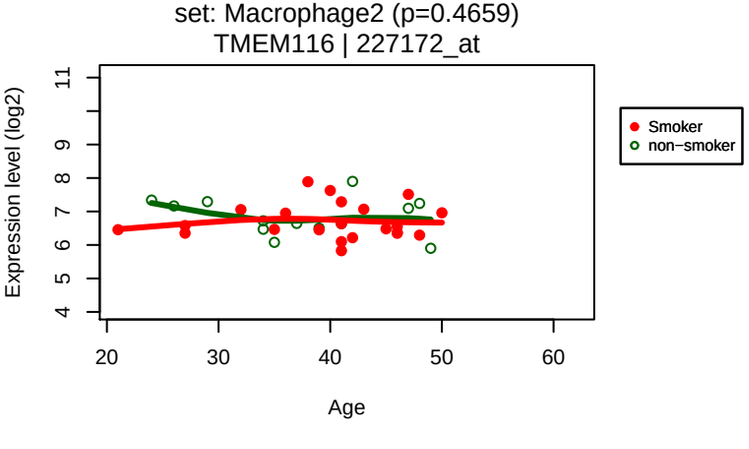

set: SmallAirway2 (p=0.0294)
TMEM116 | 227172_at
Expression level (log2)
Age
Smoker
non−smoker

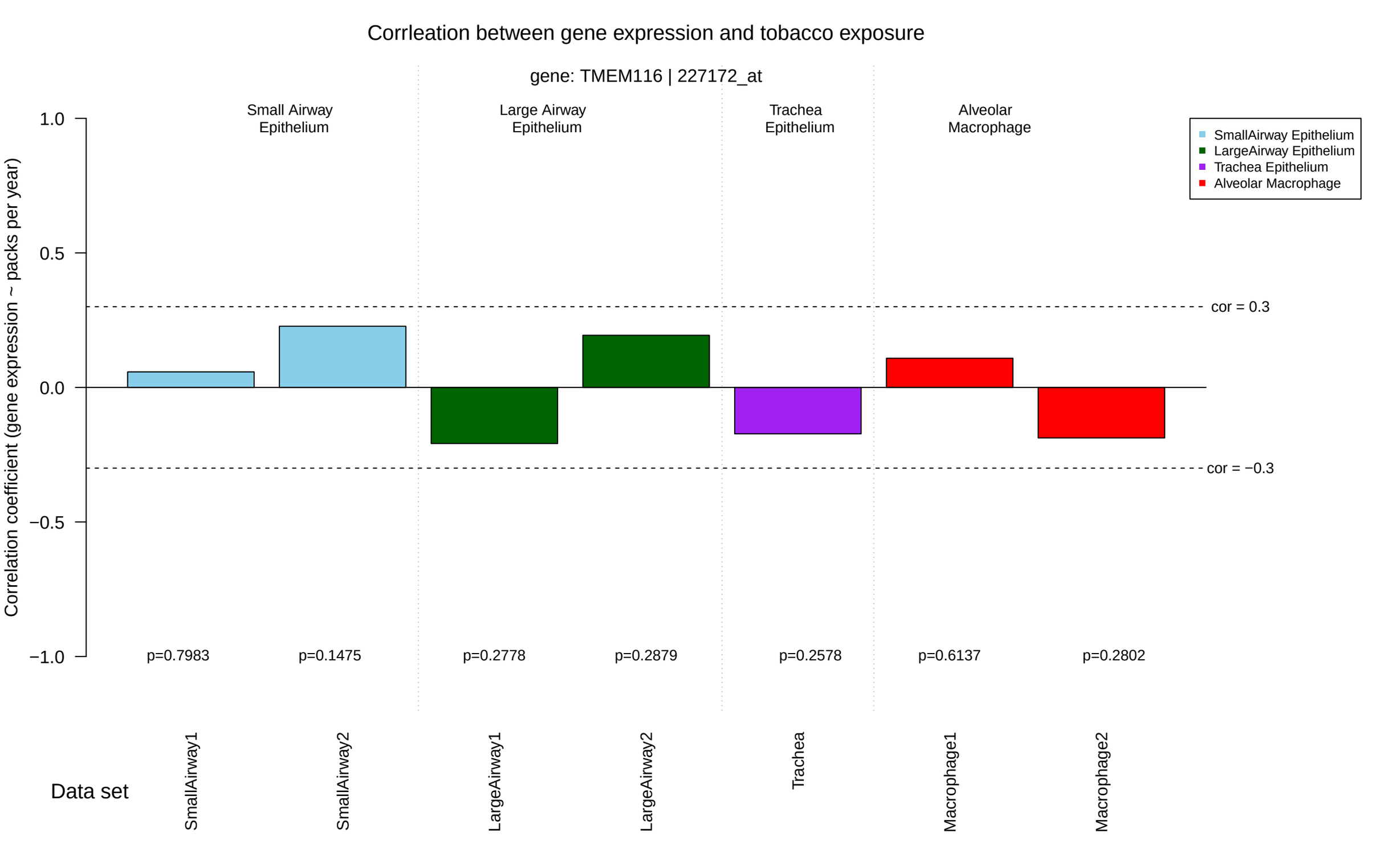

set: LargeAirway1 (p=0.1376)
TMEM116 | 227172_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7157)
TMEM116 | 227172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.178)
TMEM116 | 227172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6142)
TMEM116 | 227172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4659)
TMEM116 | 227172_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM116 | 227172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7983
p=0.1475
p=0.2778
p=0.2879
p=0.2578
p=0.6137
p=0.2802
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2